Research Methodology

Week 10 Assignment: Questions, Answers and Explanations

Week 10: Incremental versus Transformational Research, and Imagination in Research

NPTEL course noc26_ge79 (SWAYAM), 10 questions, 1 point each

The portal records no submission from you for this week, so it shows 0 on every question. The correct answers below are the portal's accepted answers. For every option the table gives a verdict and the reason, and each question ends with a short explanation of the correct answer.

Answer key at a glance

Question	Type	Correct option(s)	Portal score
1	Select all that apply	A, B, C	Not submitted
2	Single answer	A	Not submitted
3	Select all that apply	A, B, C	Not submitted
4	Select all that apply	A, B	Not submitted
5	Select all that apply	A, C	Not submitted
6	Single answer	B	Not submitted
7	Single answer	A	Not submitted
8	Single answer	C	Not submitted
9	Single answer	D	Not submitted
10	Single answer	A	Not submitted

Questions with explanations

Question 1 of 10

Students in schools typically expect

Select all that apply.

Option	Statement	Verdict	Reason
A	**Well defined questions**	**Correct**	School questions state exactly what is asked.
B	**Question paper patterns to be finalized in advance**	**Correct**	Students expect to know the paper pattern.
C	**Clarity on the "portions" for an exam**	**Correct**	The syllabus for the exam is fixed and known.
D	Questions outside the syllabus	**Wrong**	Students do not expect, and usually object to, questions outside the syllabus.

Correct answer: A and B and C.

Explanation: School exams come with clear questions, a known paper pattern and a defined syllabus. Questions from outside the syllabus are not expected.

Question 2 of 10

From the perspective of a school student the question "How long will it take to go from Chennai to Kolkata" is

Single answer.

Option	Statement	Verdict	Reason
A	**A "poorly" defined question**	**Correct**	The mode of travel, route, speed and start time are all left open.
B	Is a well defined question	**Wrong**	A well defined question would state these details.
C	Is typically how a full question is framed in an exam	**Wrong**	An exam question would give the needed details.
D	Has a unique (cut off by the portal)	**Wrong**	Different assumptions give different answers, so there is no unique answer.

Correct answer: A.

Explanation: The time depends on things the question does not say: how you travel, which route, how fast and when you start. A school question would specify them, so this one is poorly defined, like many real problems.

Note: *The portal cuts off the last option after "Has a unique"; it is presumably "Has a unique answer", which is wrong for the same reason as option B.*

Question 3 of 10

The format of exams in schools

Select all that apply.

Option	Statement	Verdict	Reason
A	**Helps you understand fundamentals**	**Correct**	Structured exams test the basics.
B	**Teaches you to breakdown a problem to parts and solve them**	**Correct**	Exam problems are solved step by step.
C	**Prepares you for a professional setting**	**Correct**	It builds discipline and problem-solving habits.
D	Teaches you to handle unexpected questions	**Wrong**	The exam format avoids unexpected questions.

Correct answer: A and B and C.

Explanation: The school exam format builds fundamentals, the habit of breaking a problem into parts and discipline for professional work. It does not teach handling unexpected questions, because the format avoids them.

Question 4 of 10

A typical professional workplace requires you to

Select all that apply.

Option	Statement	Verdict	Reason
A	**Handle open ended questions**	**Correct**	Work problems are rarely neatly framed.
B	**Solve “poorly” defined problems**	**Correct**	You must work out what is being asked.
C	Focus on syllabus	**Wrong**	A syllabus belongs to school.
D	Work within a question paper pattern	**Wrong**	A paper pattern belongs to exams.

Correct answer: A and B.

Explanation: Professional problems are open-ended and poorly defined. A syllabus and a question paper pattern belong to school.

Question 5 of 10

Learning in college is expected to

Select all that apply.

Option	Statement	Verdict	Reason
A	**Prepare you for professional life**	**Correct**	College is the bridge between school and work.
B	Focus only on well-defined problems	**Wrong**	The word "only" makes it wrong; college must go beyond that.
C	**Teach you to handle open ended problems**	**Correct**	It prepares you for open-ended work problems.
D	Follow the same approach used in school	**Wrong**	College is expected to go beyond the school approach.

Correct answer: A and C.

Explanation: College should prepare you for professional life, and it does so by teaching you to handle open-ended problems. Repeating the school approach would not do that.

Question 6 of 10

Applying a known theory to one additional new situation is most likely

Single answer.

Option	Statement	Verdict	Reason
A	Breakthrough research	**Wrong**	A breakthrough changes the field; this extends it slightly.
B	**Incremental research**	**Correct**	It extends existing knowledge by a small step.
C	Transformational research	**Wrong**	A known theory in one more case does not transform the field.
D	Important contribution to research	**Wrong**	It is too vague and not the type of research asked for.

Correct answer: B.

Explanation: Applying an existing theory to one more situation adds a small step to what is known, which is incremental research.

Question 7 of 10

Use of a series of Experimental techniques to study the same phenomenon across different samples is typically an example of:

Single answer.

Option	Statement	Verdict	Reason
A	**Systematic research**	**Correct**	A planned, orderly series of studies of one phenomenon is systematic research.
B	Transformational research	**Wrong**	It does not change the direction of the field.
C	Incremental research	**Wrong**	The key calls this organised programme systematic, not incremental.
D	Breakthrough research	**Wrong**	It does not produce a sudden major advance.

Correct answer: A.

Explanation: Studying one phenomenon with a planned series of techniques across different samples is an orderly, methodical programme, which is systematic research.

Question 8 of 10

In the development of phones, which of the following can be considered transformational

Single answer.

Option	Statement	Verdict	Reason
A	First deployment of the landline	**Wrong**	An early step; the key does not count it as the transformation asked for.
B	First deployment of the cordless phone	**Wrong**	It freed the handset only within the home.
C	**The improvement from cordless phone to mobile phone**	**Correct**	The mobile phone removed the dependence on a fixed place.
D	Making the first-generation mobile phones smaller and more colorful	**Wrong**	This only refines an existing product.

Correct answer: C.

Explanation: The jump from the cordless phone to the mobile phone changed how people use a phone, since it goes with you everywhere. Making phones smaller or more colourful only refines it.

Question 9 of 10

In the development of bicycles, which of the following can be considered transformational

Single answer.

Option	Statement	Verdict	Reason
A	Improvement from wooden seat to cushioned seat	**Wrong**	A comfort improvement.
B	Making the seat more ergonomic	**Wrong**	A comfort improvement.
C	Adding a carrier	**Wrong**	A convenience, not a change in capability.
D	**Adding gears to a bicycle**	**Correct**	Gears change what the bicycle can do: climb slopes and cover long distances with the same effort.

Correct answer: D.

Explanation: Gears change the capability of the bicycle. A better seat or a carrier improves comfort or convenience without changing what the bicycle can do.

Question 10 of 10

In the development of airplanes, we can associate imagination with

Single answer.

Option	Statement	Verdict	Reason
A	**Designing fixed wings even though birds have flapping wings**	**Correct**	It goes beyond copying nature.
B	Adding seats inside an airplane	**Wrong**	A practical refinement.
C	Creating a cargo version of the airplane	**Wrong**	An extension of an existing design.
D	Adding more engines to the airplane	**Wrong**	An engineering improvement.

Correct answer: A.

Explanation: Imagination means going beyond what nature or the current solution shows. Choosing fixed wings with engines for thrust, instead of copying the flapping wings of birds, was such a leap.